Section 5
Vestibular & Somatosensory System

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COGS 17

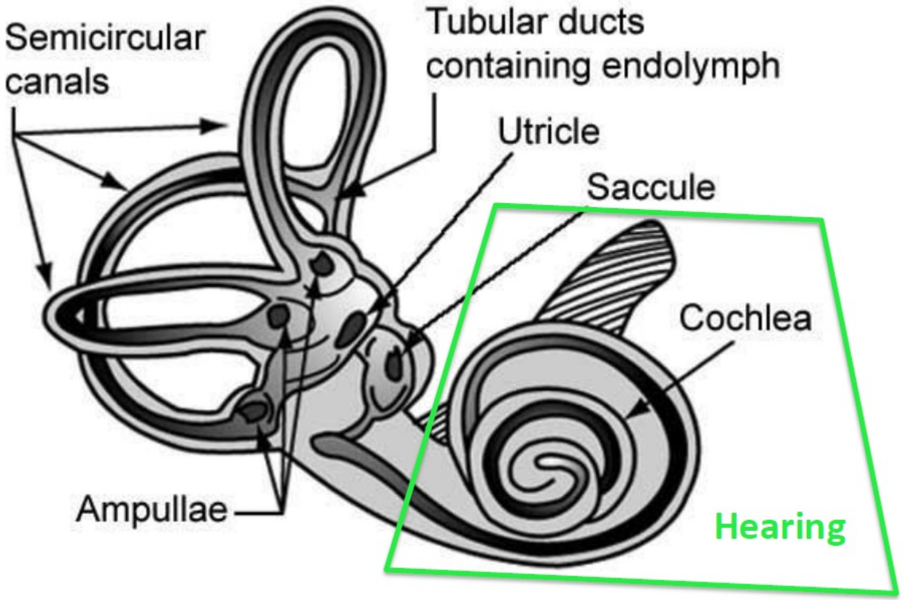
11/07/25

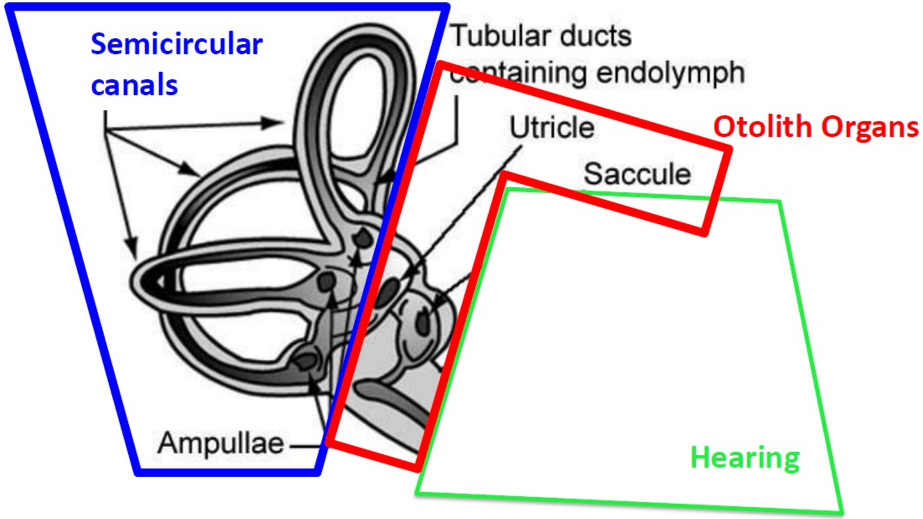
UC San Diego

Announcements

- Homework for Friday, Nov 7th is **STILL DUE**
- Midterm 2 is **STILL Next TUE**
- **No TA discussion section** next week!

Vestibular System

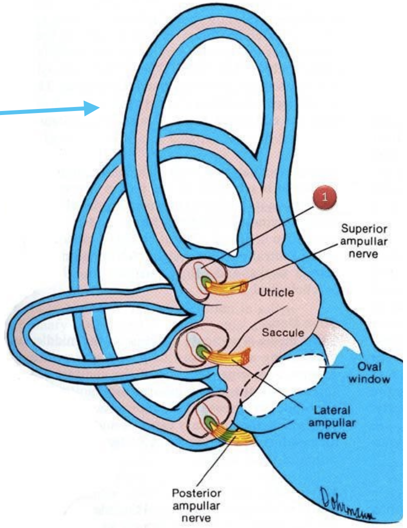


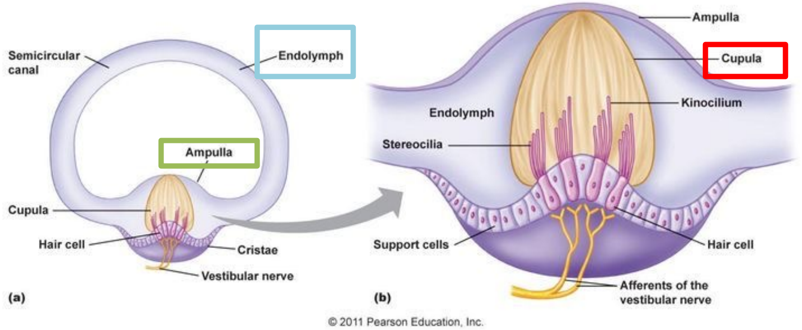


Vestibular System

- 2 structures: **Semi-circular canals (for ROTATION) & Otolith Organs (HEAD TILT) →** provide movement and balance info

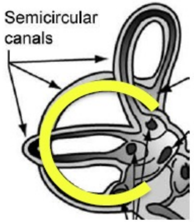
- **Semi-circular canals**
  - 3 looped tubes orientated to different orthogonal planes (XYZ-planes) - filled with K+ rich **endolymph fluid**
  - **Hair cell (vestibular receptor cells):** in a chamber (Ampulla) at the base of the semi-circular canals, embedded in a gelatinous cap (Cupula), its cilia are bent by flow of endolymph in Ampulla
  - When the head rotates (start of rotation), fluid in ampulla lags behind movement
  - When head movement stops (end of rotation), the fluid overshoots movement
  - This motion deforms the cilia of the hair cells and alters NT release

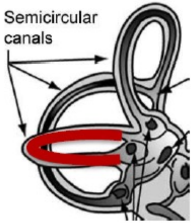


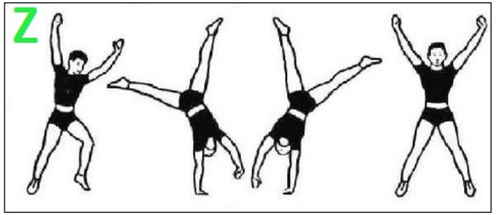


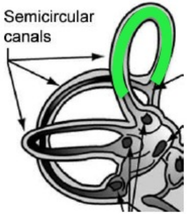
Vestibular System

Semi-Circular Canals - ROTATION









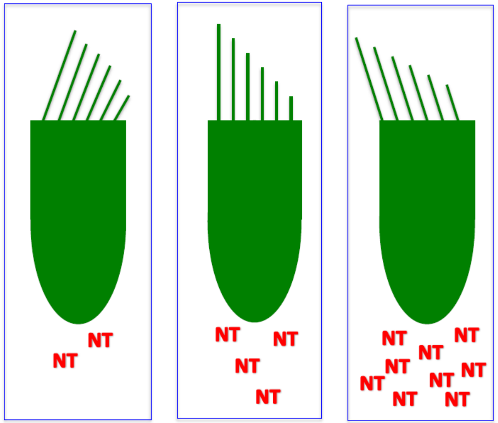
Vestibular Hair Cells

- Similar with Hair Cells in Auditory system, except… that Vestibular HC has spontaneous firing, in the absence of input
  = When no rotation, hair cells release a base rate of NTs

- Rotation of endolymph fluid causes the cilia of hair cells to bend
  - K+ in the endolymph fluid enters the hair cells to open Ca^2+ gates which facilitates NT release
  - If the cilia bends towards the **tallest, more** NT is released (opposite: **shortest → less** NT)
  - Graded potentials based on the magnitude of rotation/bend

Bend cillia one way,

LESS neurotransmitter is released



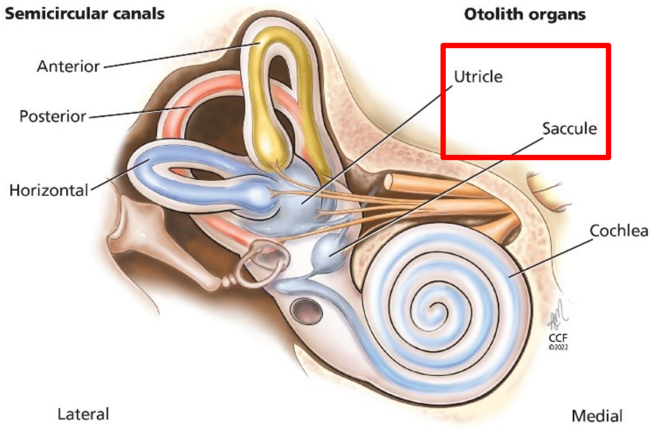
Base rate

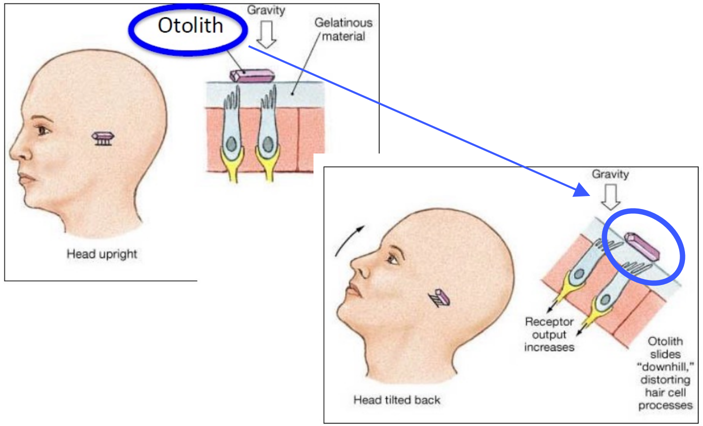
Bend cillia other way,

MORE neurotransmitter is released

Otolith Organs-1

- Detects **<u>changes</u>** <u>in head tilt</u> relative to the body
- Consists of the 2 chambers – Utricle and Saccule (together form the **Macula**): HCs line the walls of these Endolymph-filled chambers
- **“Ear Stones” (Otoliths)**: Calcium-Chloride crystals sit in gelatinous material in which HCs are embedded
- When head is upright: Base firing rate
- Whichever angle you tilt your head, some Otoliths weigh down the cillia of some HCs (bend one way, fire more, bend other way, fire less)
- Note that Acceleration & Deceleration has the same effect as tilt backward & tilt forward, respectively
- Otolith organs detect the start/stop of locomotion as well as head tilt

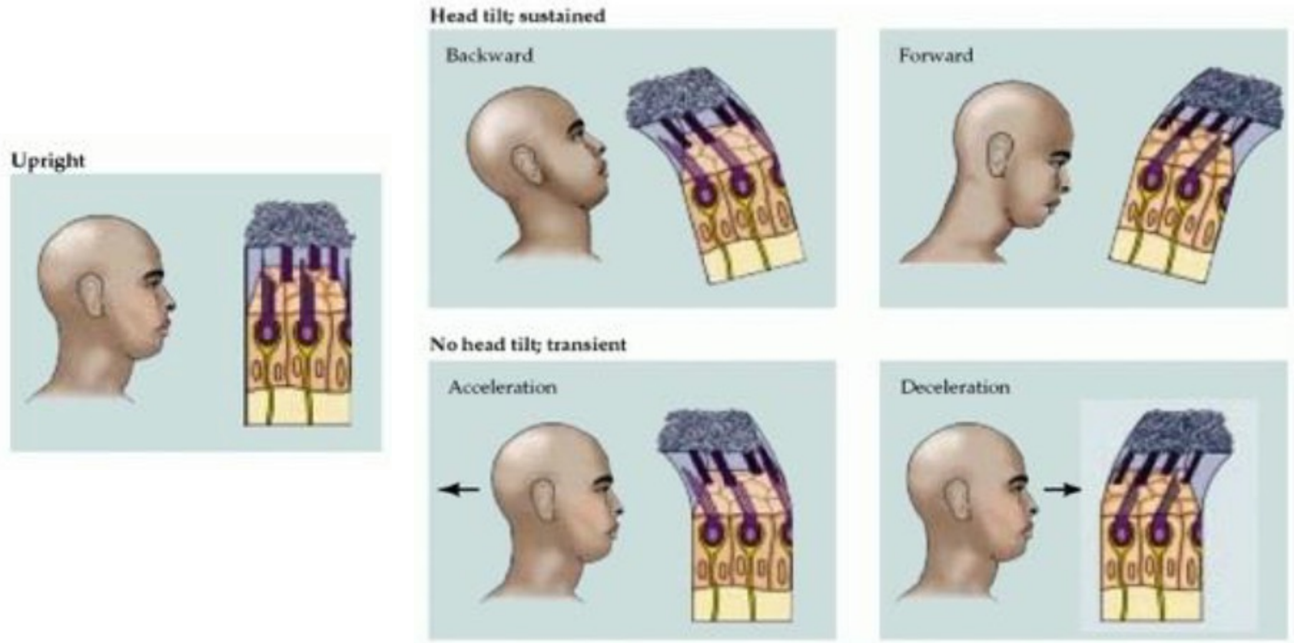




Otolith Organs-2

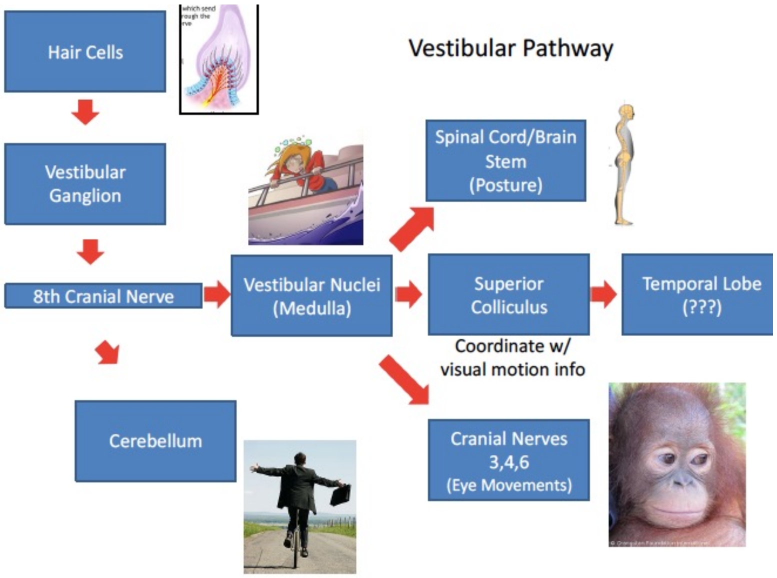
- Note that Acceleration & Deceleration has the same effect as tilt backward & tilt forward, respectively
- Otolith organs detect the start/stop of locomotion as well as head tilt

→ Vestibular System only detects CHANGE



Vestibular Pathway

- HCs cause NT release which synapses onto vestibular ganglions
- Vestibular ganglions project to 8th cranial nerve
- From the 8th cranial nerve splits:
  - Cerebellum: Direct connection for maintaining balance
  - Vestibular nuclei (medulla): plays an essential role in maintaining equilibrium, posture, head position, etc. (disruption leads to nausea)
- Medulla projects to 1) spinal cord/brain stem (regulate posture), 2) superior colliculus (coordinate with vision), and 3) cranial nerves 3,4, and 6 (eye movements so you can keep focus while moving)
- Higher pathways are still areas of research
- **Motion sickness:** When vestibular and visual systems are not coordinated or improperly integrated.

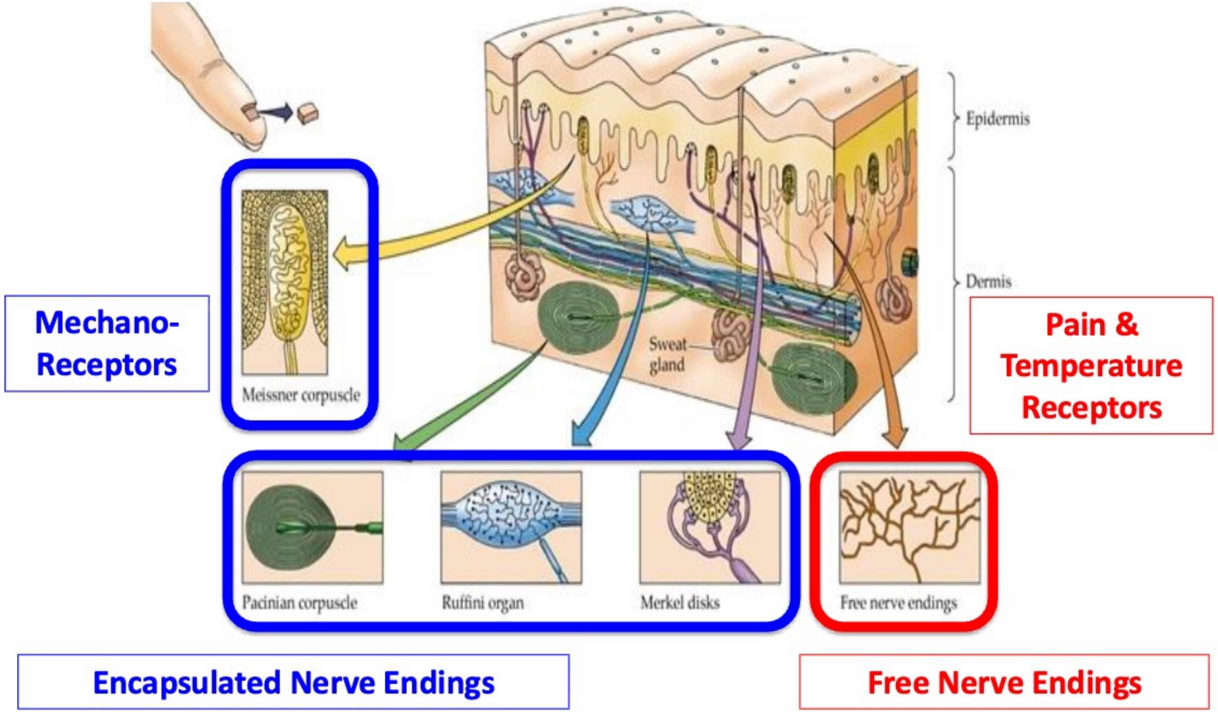


Vestibular Pathway

Somatosensory Receptors-1

Two basic types: Free Nerve Endings & Encapsulated Nerve Endings

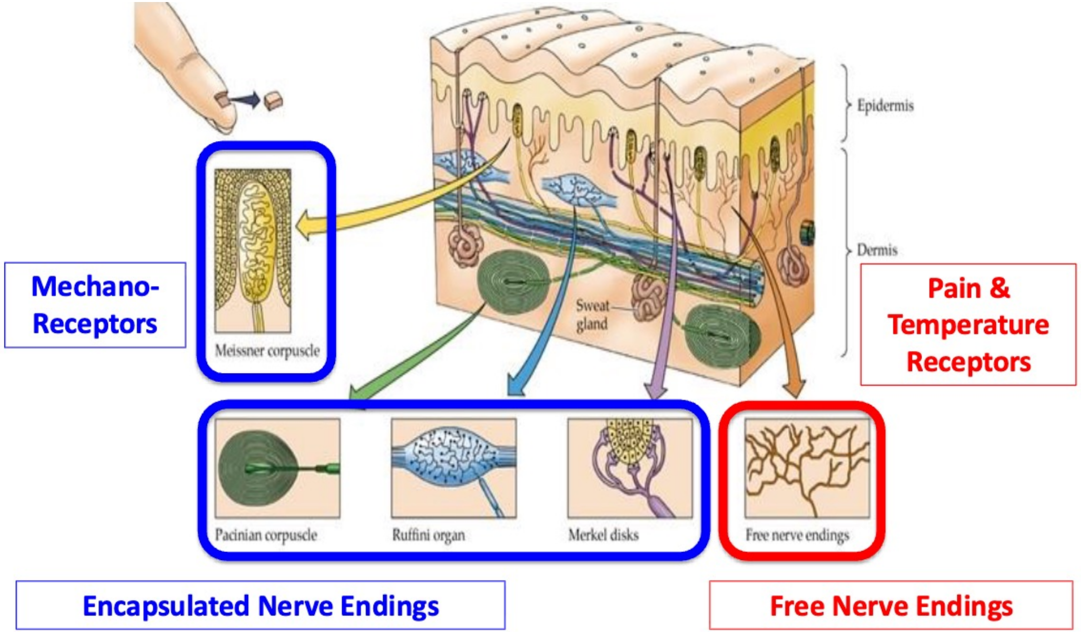
Both responds to Deformation (or other physical impact on cell), leading to Action Potentials in these Receptor cells



Somatosensory Receptors-1

1. Free Nerve Endings:

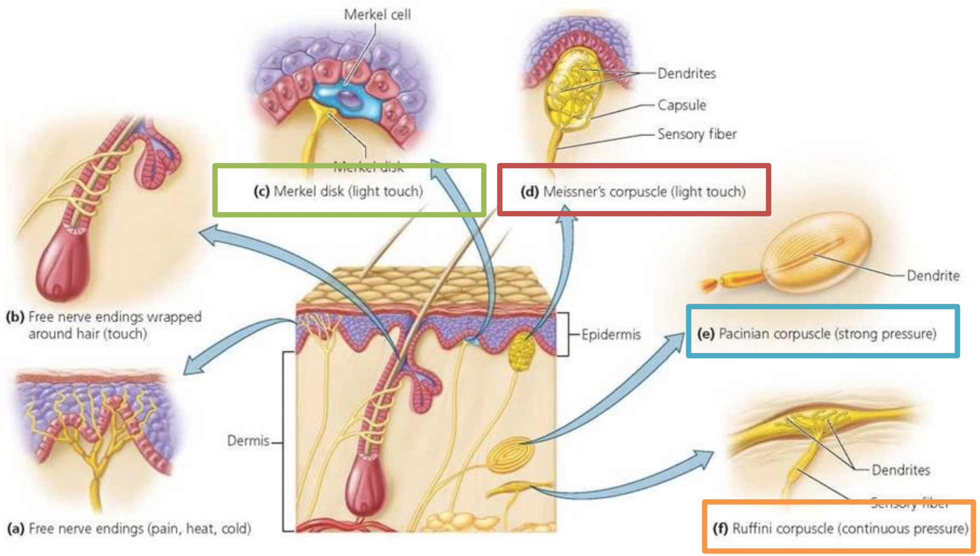
- **Nociceptors:** detects pain, itchiness, extremes of temp
- **Thermoreceptors:** detect Temperature
- **Follicle Receptors:** wraps around hair follicles to detect movement

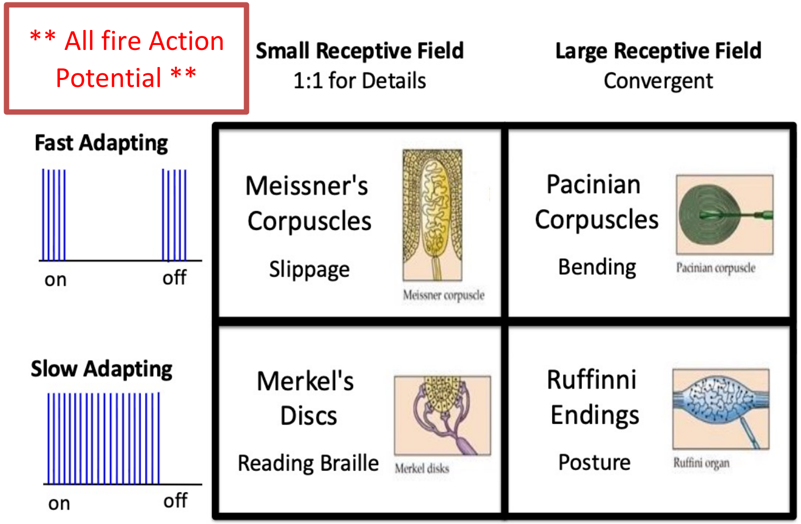


Somatosensory Receptors-3

2. Encapsulated Never Endings (aka Mechano-Receptors)

- **Proprioception:** Internal muscle & organ movement
- **Touch:**
    - **Meissner: Small** receptive Fields & **Fast** adapting, responds best to rapid changes (slippage)
    - **Merkel: Small** receptive Fields & **Slow** adapting, responds best to detail discrimination (Braille, textures)
    - **Pacinian: Large** receptive Fields & **Fast** adapting, responds best to large scale changes (bending)
    - **Ruffinni: Large** receptive Fields & **Slow** adapting, responds best to sustained large scale events (sitting)





Across Fiber Coding

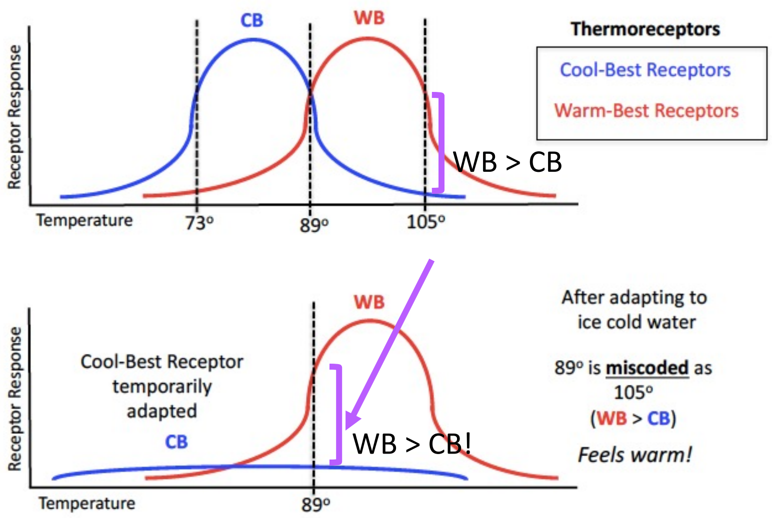
- Relative Proportion of activity across fiber codes for a given temperature

- Two types of Thermoreceptors:
  - Cool-best receptors (CB)
  - Warm-best receptors (WB)

- Each are selectively tuned to prefer lower or higher temps

- Given the following:
  - WB = CB = "Physiological Zero" 89 F
  - WB < CB = "Cold feeling" 73 F
  - WB > CB = "Warm feeling" 105 F

- Selective Adaptation: WB or CB is adapted and "exhausted"
  - After adapting to cold water, 89 F will feel like 105 F due to miscoding of the CB receptor being exhausted
  - After adapting to warm water, 89 F will feel like 73 F due to miscoding of the WB receptor being exhausted

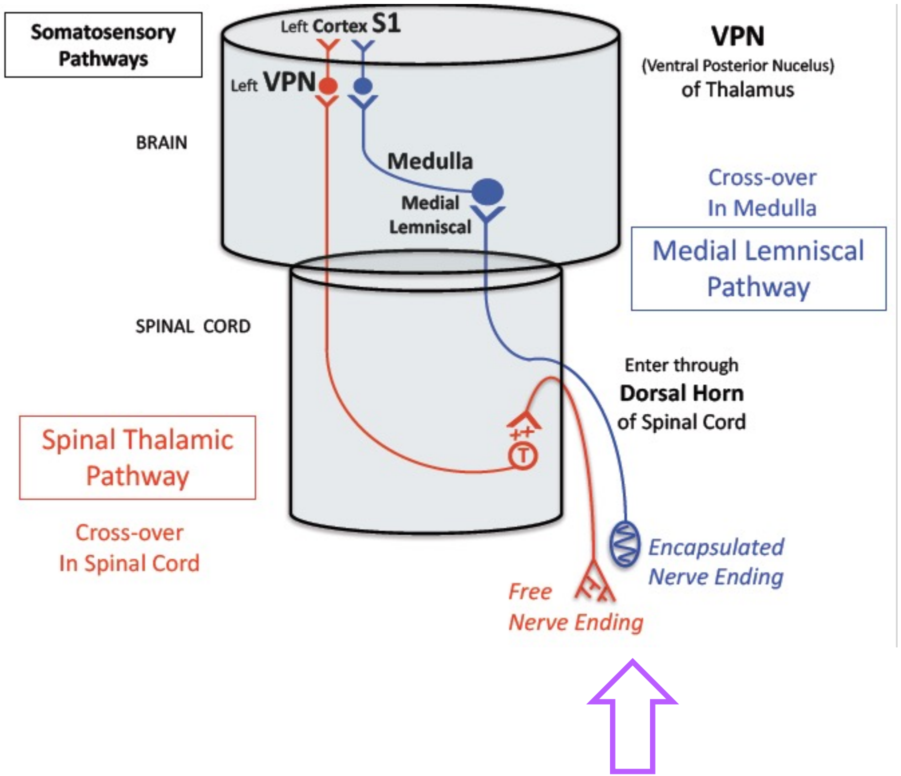


Somatosensory Pathways-1

- Different pathways for Free Nerve Endings and Encapsulated Nerve Endings

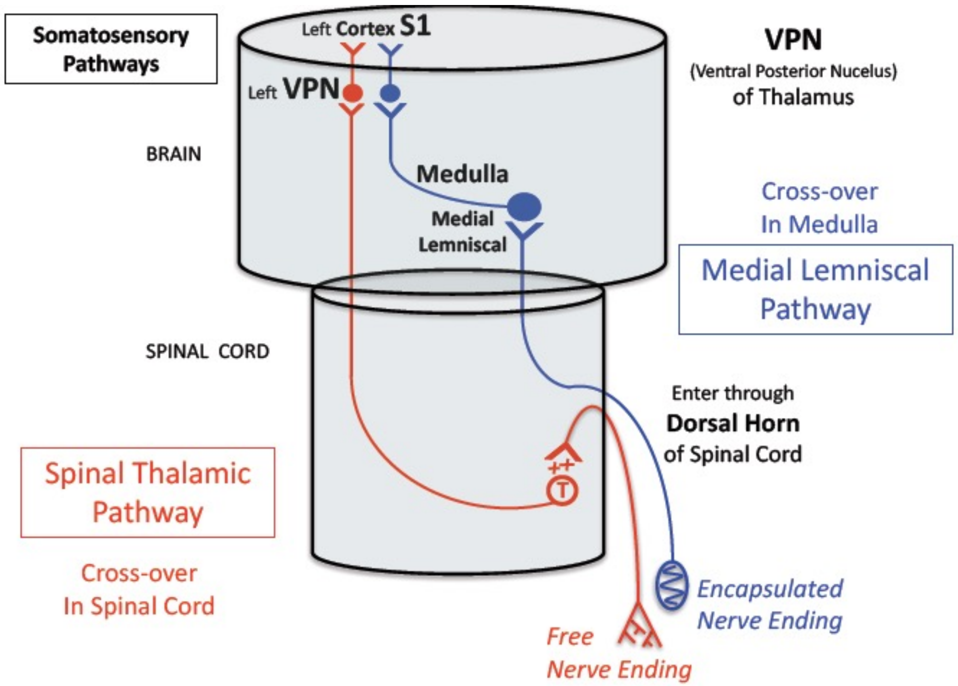
- **Spinal Thalamic Pathway**
  - Free Nerve Endings enter Dorsal Root of Spinal Cord and synapse there
  - "Second-Order" neurons cross over in spinal cord → ascend to synapse in contralateral ventral posteromedial nucleus (VPN) of thalamus (relative to the origin of the signal)
  - Cross-over happens within the spinal cord!

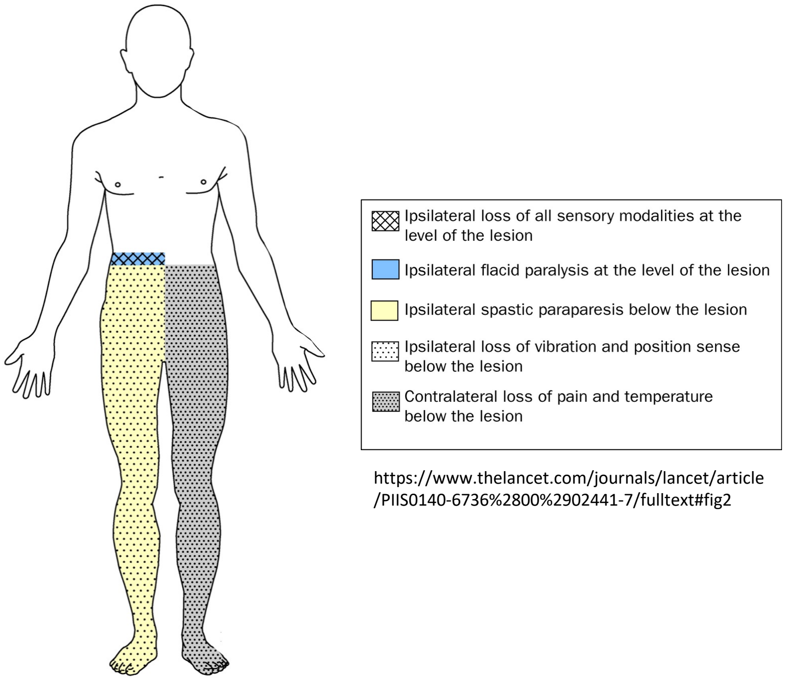
- **Medial Lemniscal Pathway**
  - Encapsulated Nerve Endings enters Dorsal root, (one collateral of axon synapses there)
  - Main fiber ascends on ipsi-lateral side of spinal cord & medulla
  - "Second-order" cells cross-over in the Brain Stem (Medial Lemniscal tract) to synapse in contra-lateral VPN



Somatosensory Pathways-2

- **Brown-Sequard Syndrome:** damage to only one side of the spinal column (e.g., the right)
  - → reduction/loss of touch and position on ipsilateral side (right) below the point of injury
  - → reduction/loss of temperature and pain detection on the contra-lateral side (left) below the point of injury





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Somatosensory Pathways in the Brain

In Both Pathways: First major synapse occurs at place of cross-over

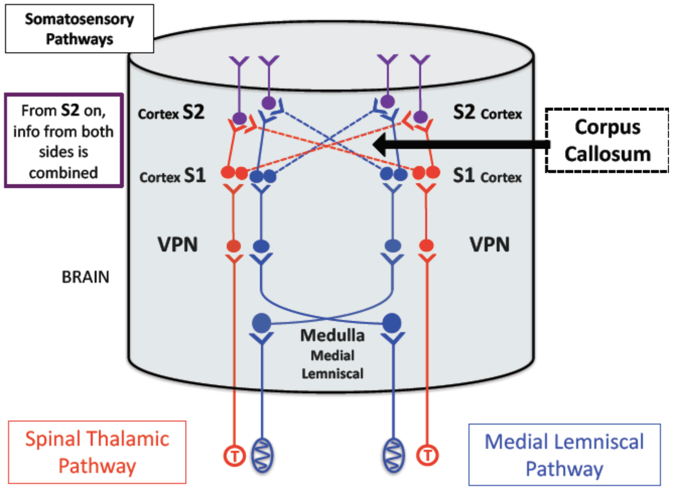
From VPN (thalamus), message continues along contra-lateral side to S1, and then S2 (both in Parietal Cortex)

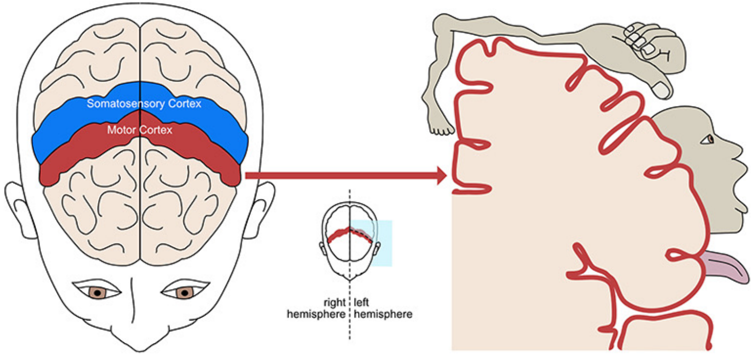
After S1, Corpus Callosum exchanges info, so S2 reacts to both sides (although shows dominance for contra-lateral side)

Somatosensory Cortex

- Post-Central Gyrus of Parietal Lobe, just posterior to Central Sulcus that separates Parietal/ Frontal Lobes
- Contains a topological map of body surfaces
- Magnification Factor of Penfield Map: areas with small receptive fields have a high density of high acuity receptors takes up a disproportionately large area of the cortical map

→ Homunculus (large hands, mouth vs. small torso)





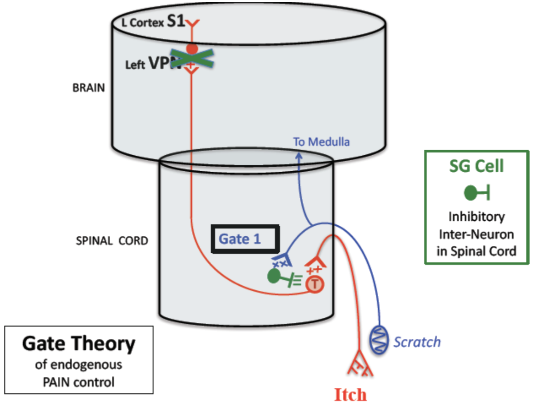
https://kids.frontiersin.org/articles/10.3389/frym.2022.750301

Gate Theory – Pain Control

… via negative feedback from brain or body

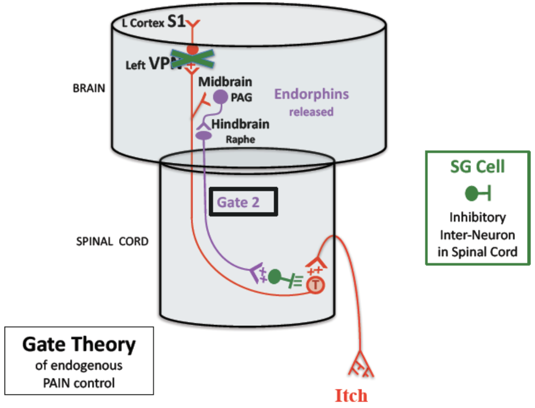
- Basic mechanism: Nociceptors release Substance P in the Spinal Cord → stimulate T cells to send message up to the Brain

  … but T cell activity can be inhibited by ”closing the gate”



1. By scratching next to a bite, SG cells activate and inhibit T cells to “close the gate”

2. Periaqueductal Gray Area (PAG) in the midbrain
- Releases inhibitory endorphins → Inhibits inhibition in raphe system (brain stem)
- Raphe system sends excitatory NT to spinal cord to stimulate SG cells to inhibit T cells from reacting to substance P



3. Within brain, some cells that release Substance P have NT receptor sites on their Terminals that respond to inhibiting Endorphins
- So, via Axo-Axonal connections, Endorphins directly inhibit Presynaptic release of Substance P in brain

Control of Movement

Muscles

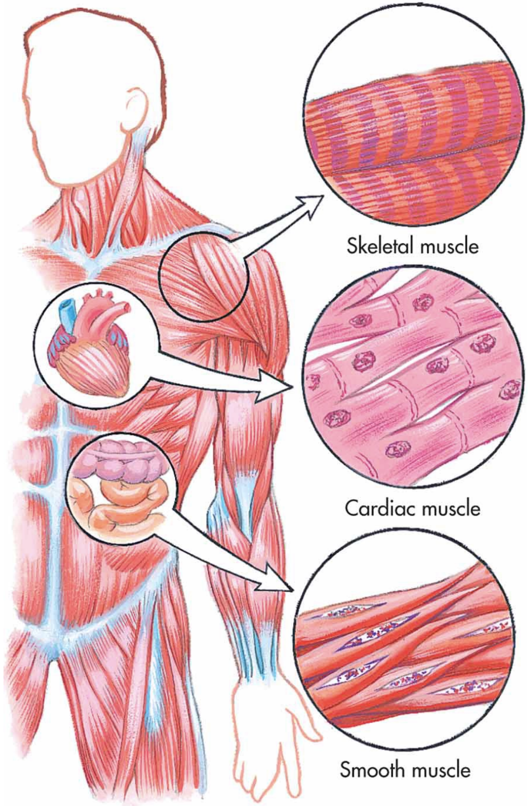
Three types of Muscles:

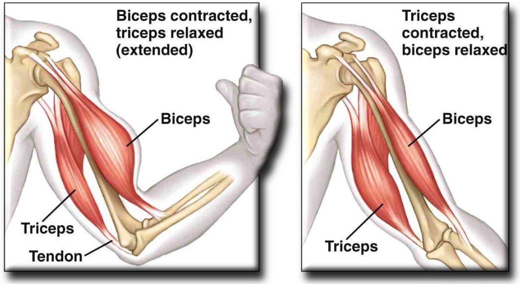
1. **Striate (Skeletal, Facial):** Connects tendons to bones for voluntary movements

2. **Cardiac (Heart):** has endogenous rhythms of activity that is modified by neurons

3. **Smooth (Internal Organs):** Can sustain contraction, mostly autonomically controlled

- **Skeletal Muscles:** come in "Antagonistic Pairs"
    - **Flexor** moves bone toward body
    - **Extensor** move same bone from body





Neuro-Muscular Junction-1

- primarily involving Striate Muscles
- Motor Neurons (Alpha Neurons) release Acytelcholine (Ach) onto muscle fibers
- (like in a neuron…) Muscle fiber response is all/nothing depolarization
    - Na+ and then K+ gates open/close
    - Ca++ enters muscle cells → triggers energy-requiring process that causes muscle contraction
    - afterward, Ca++ is actively pumped out, and a Na+/K+ Pump repolarizes fiber for next contraction

BUT instead of causing NT release, Ca++ activates Sarcomeres (contractile unit in Skeletal Muscles) to contract the muscle

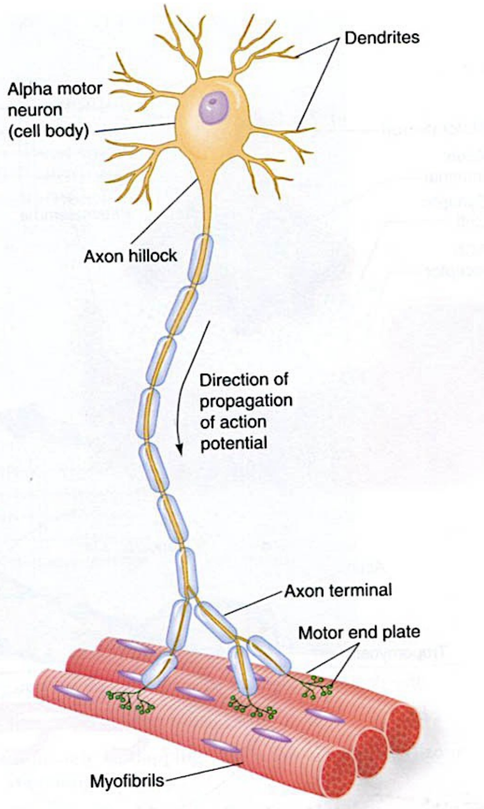


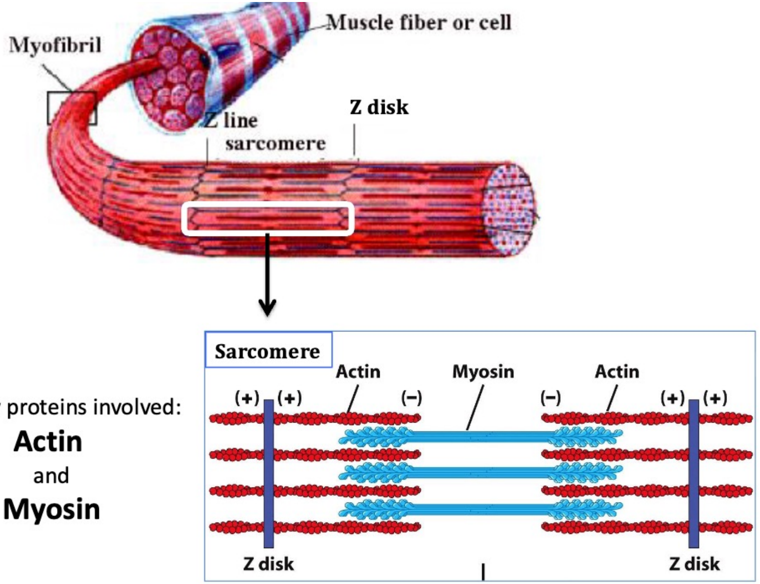
Figure 1.6 Motor units include the α-motor neurons and the muscle fibers they innervate.

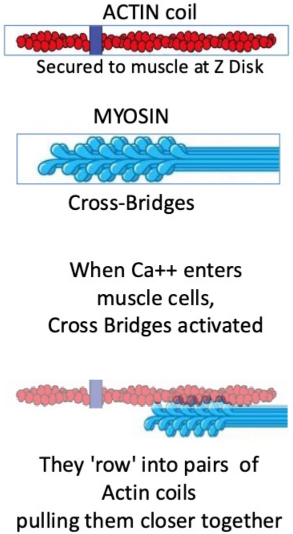
Neuro-Muscular Junction-2

- Within each fiber is a string of Sarcomeres (the contractile units) each consisting of…
  - **Actin:** thin protein filament, anchored to muscle
  - **Myosin**: thick protein filament with ”cross-bridge” heads

→ Contraction = Myosin Cross Bridges hook into (grab) coiled Actin, bend to tighten coil, release, repeat

Two key proteins involved:
Actin
and
Myosin





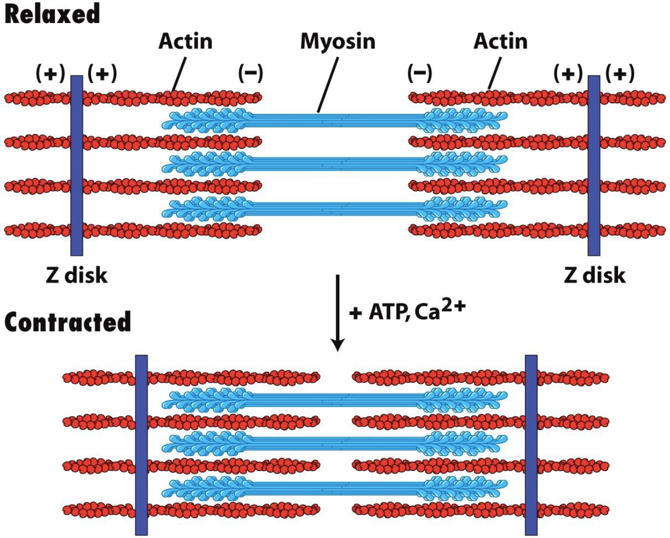


Figure 17-30
Molecular Cell Biology, Sixth Edition
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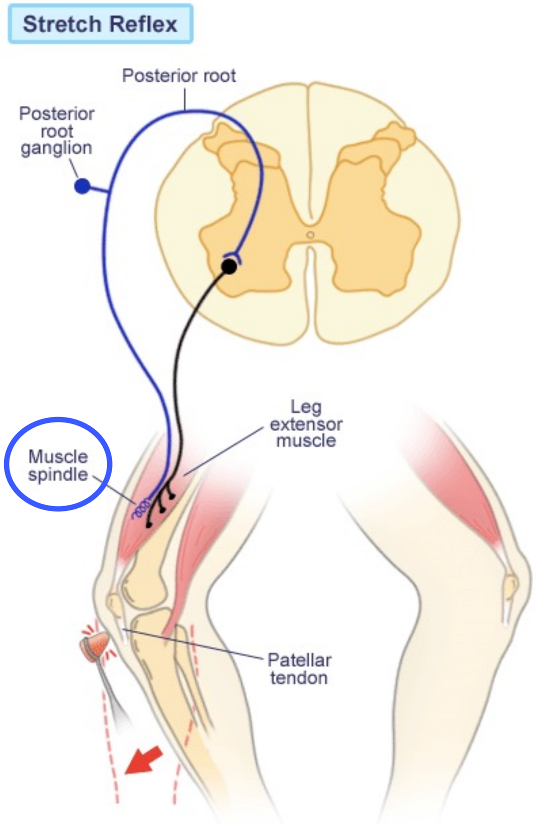
Spinal Reflexes

1. **Stretch Reflex:** The only mono-synaptic reflex (one synapse)
   - Muscle Spindle (A Proprioceptor) in muscle detects passive stretch of muscle
   - Axon of muscle spindle to spinal cord → excites Motor neuron back to same muscle → muscle contracts to counter the stretch

- **Golgi Reflex**
  - Proprioceptors called Golgi Tendon Organs in tendons detect intensity of muscle contraction
  - If contraction is too strong (threatens to tear muscle apart) sends signal to Interneurons in Spinal Cord
    that inhibit the Motor Neurons causing that contraction, lessening their rate of firing
    - Note that since striate muscles come in antagonistic pairs, inhibiting a given flexor usually also involves
    a parallel circuit to excite its paired extensor (and inhibiting extensor involves exciting its flexor)



Spinal Reflexes

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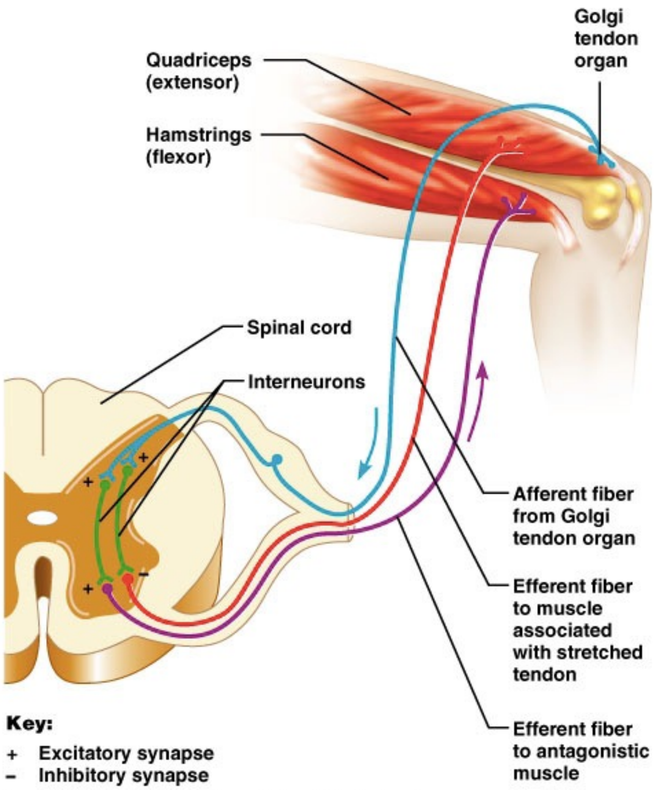
   - Muscle Spindle (Proprioceptors) in muscle detects passive stretch of muscle
   - Axon of muscle spindle to spinal cord → excites Motor neuron back to same muscle → muscle contracts to counter the stretch

2. Golgi Reflex

- Golgi Tendon Organs (Proprioceptors) in tendons detect intensity of muscle contraction
- If contraction is too strong (threatens to tear muscle apart), sends signal to <u>Inhibitory</u> Interneurons in Spinal Cord that inhibits Motor Neuron to original muscle → reducing contraction

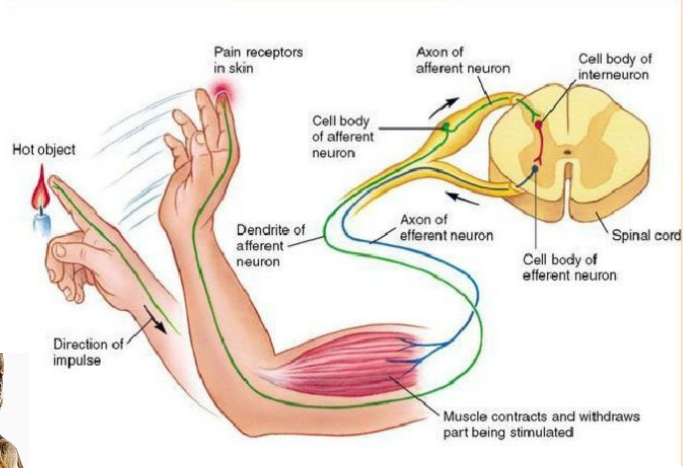
- Golgi Tendon Organ also activates an <u>Excitatory</u> Interneuron in Spinal Cord that excites Motor Neuron to Antagonistic muscle → increasing its contraction / decreases original's contraction



Spinal Reflexes

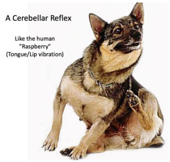
3. Pain Withdrawal Reflex (Touch a hot stove → jerk hand away)

- Stimulated Nociceptors signal Interneurons in Spinal Cord to excite Motor Neurons that synapse back onto relevant Flexor muscles to move body part away from noxious stimulus
- Signals sent along myelinated Motor Neurons reach muscle before Pain signal even reaches brain



4. Scratch Reflex (dog's rhythmic scratch with hind leg)

- Rate is relatively fixed, mediated by spinal cord
- such oscillator circuits might be produced by Central Pattern Generators in Cord, Cerebellum
- In humans, probably involved many learned "motor programs" (dance, speech, writing…)



5. Infant Reflexes

- "Rooting": touch to cheek → head turn & infant suckles
- "Bapkin": press palms → fingers grasp & mouth opens

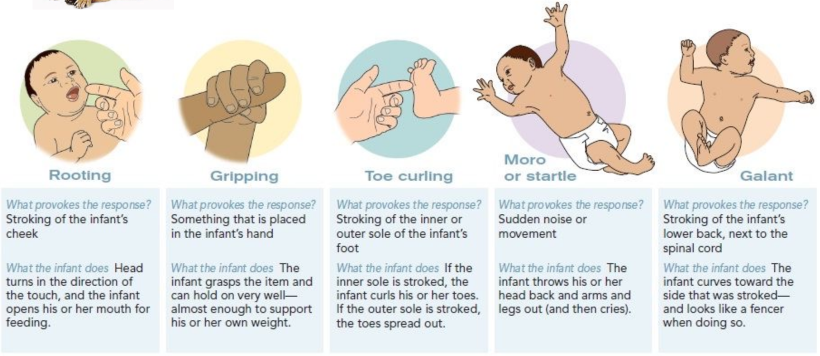
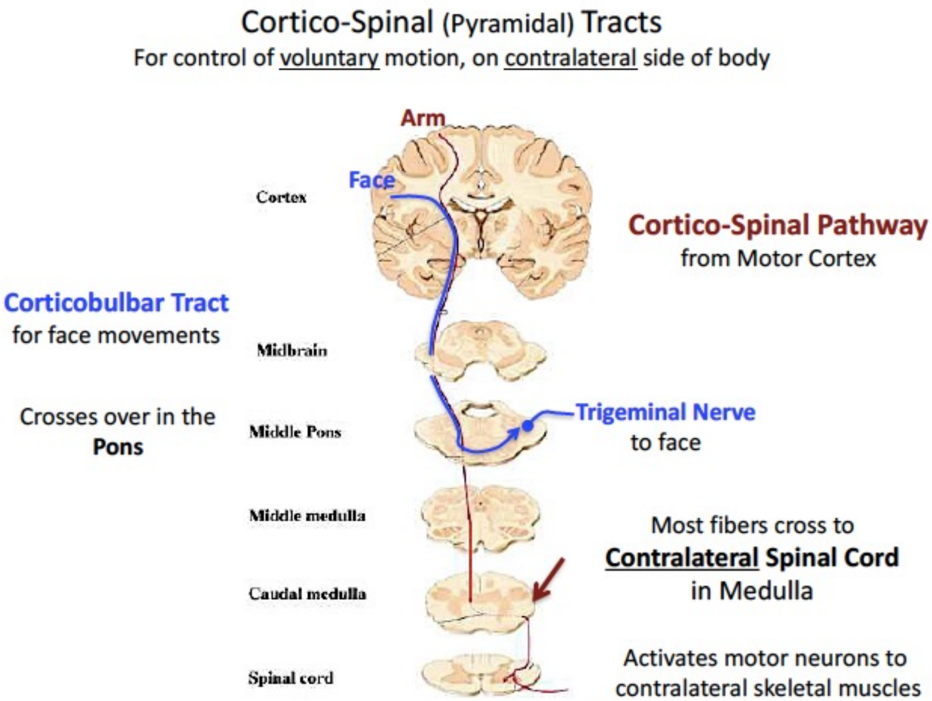


FIGURE 9.2 **Some Infant Reflexes** Infants are born with a number of reflexes to get them through life, and they are incredibly cute when they perform them. These reflexes disappear as infants mature.

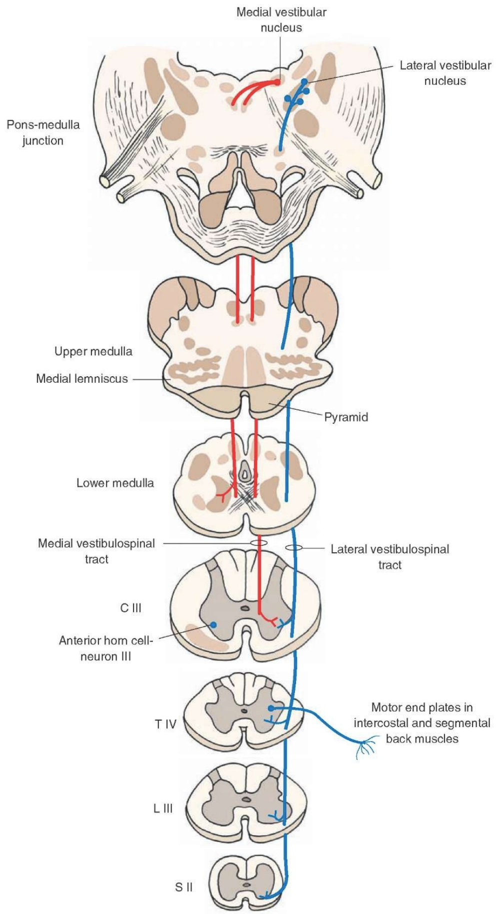
Cortico-Spinal (Pyramidal) Tracts

- For voluntary movement control on contra-lateral side of the body (Left hemisphere controls Right side of body, etc.)
- From large pyramidal cells from Motor Cortex
- crossover at Pyramids of Medulla
- synapse in Spinal Cord (onto Inter-Neurons or Motor Neurons)
    - Some synapse first at Red Nucleus (e.g. integrate w/Vestibular & Cerebellar info)
- Fast, myelinated tracts, esp. for precise control of peripheral movements (e.g. hands, fingers, limbs, + face)
- Also includes to face, cross over in Pons, synapse on cranial (Trigeminal) nerve to face



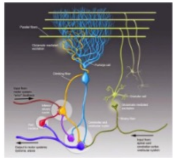
Ventro-Medial Tracts

- Mainly for Bi-Lateral & Ipsi-lateral Midline control (central body)
- Includes circuits for WALKING, since two sides must be in tight coordination
- Multiple sub-paths, originating from sub-cortex, most synapse in Spinal Cord, esp on Inter-Neurons
- Many make multiple connections in Tectum, Vestibular Nucleus, Reticular Formation, integrating with sensory & arousal systems

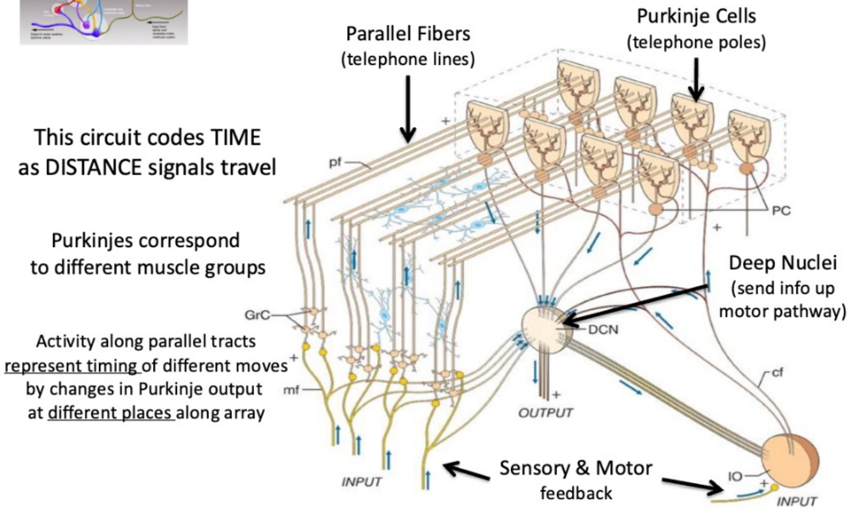


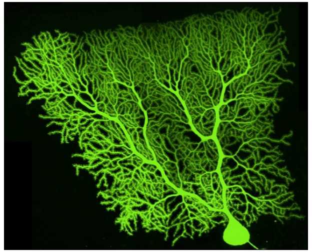
Cerebellum "The Little Brain"

- For rapid, well coordinated movements that require precise timing and control
- Includes both programmed (e.g., saccades) and learned behaviors
- Depends on real-time sensory-motor feedback, Important for attention shifting
- Composed of Purkinje cells
  - Most developed dendritic branches of any cells in the body
  - Communicates to deep nuclei to command motor nuclei in the brain
- Parallel fibers wrap along Purkinje cells
- Particularly sensitive to alcohol poisoning



Cerebellar Circuits



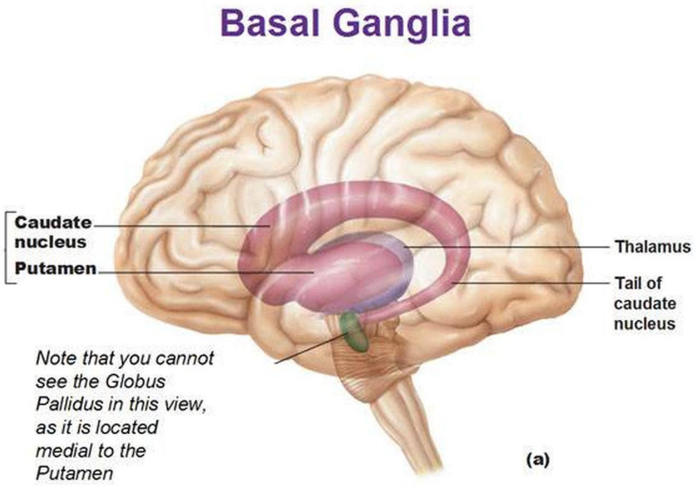


Basal Ganglia-1

- Circuit within the brain that involves multiple subcortical structures
  - **Caudate, Putamen, Globus Pallidus, and Claustrum**

- Organizes Behavior, esp learned, task-based sequences

- Caudate Nuc + Putamen (aka “Striatum”): mainly receive sensorimotor from Thalamus, Tegmentum, & Cortex
- Globus Pallidus: sends output up to Motor Cortex via Thalamus, and down via Red Nucleus to Cerebellum & Cord
- Claustrum: just sub-cortical to Insula, connects w/Cortex, esp Frontal & Sensory (Topo map of Sensory Cortex!)



Basal Ganglia-2

Not only motor…

- Direction and amplitude of slow, smooth-changing, voluntary movements (e.g., posture, walking)
- Maybe implicated in automating complex sequential processes (e.g., driving)

Pathologies:

- overactive links with PFC → Obsessive Compulsive Disorder (OCD), Attention Deficit Disorder

- Parkinson’s Disease
  - Symptoms: rigidity, tremors, difficulty in initiating/stopping movement, memory & other cognitive deficits
  - Primarily from degeneration of Dopaminergic axons from especially Substantia Nigra (Tegmentum) to Striatum
  - Result: Increased inhibition from Globus Pallidus to Thalamus, decreasing Thalamic excitation of Cortex
  - Treated primarily by L-dopa, Dopamine precursor that crosses Blood-Brain barrier

https://www.researchgate.net/publication/342612069_Early_Detection_of_Parkinson's_Disease_-Simulation_and_Assessment?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Il9kaXJlY3QiLCJwYWdlIjoiX2RpcmVjdCJ9fQ

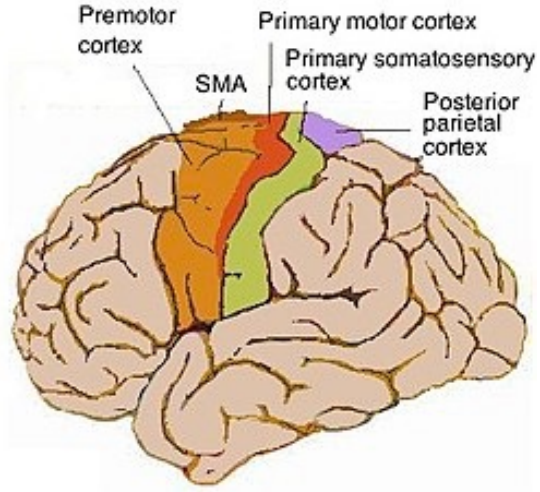
Motor Cortex

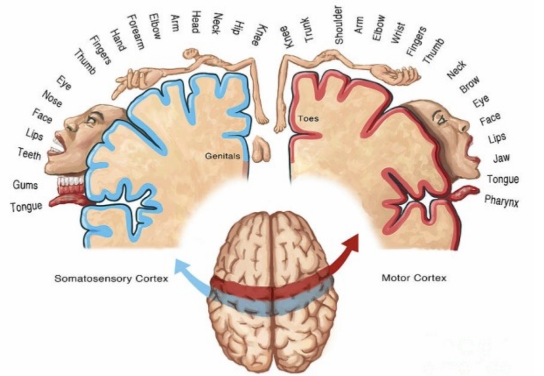
Primary Motor Cortex (M1)

- Includes topological "Map" of body; Receives from corresponding map in nearby Somatosensory Cortex
- No direct connection to muscles, but send commands to Motor Neurons in Brain Stem and Spinal Cord

Secondary Motor Cortex (Planning movement)

- <u>Premotor Cortex</u> (anterior to M1)
    - Active during "preparation to move", just preceding M1 activity
    - Mirror Cell System w/Visio-Spatial Parietal, responds to seeing own, or other's, hands doing familiar tasks
    - Also includes much of Broca's Area, involved in production of grammatical speech
- <u>Supplementary Motor Cortex</u> (anterior to M1 & dorsal to Premotor Cortex)
    - Like Premotor, active during preparation, but especially for rapid sequences of movements
    - Receives from Parietal Cortex including somato- (esp. proprioceptive) & posterior visuo-spatial maps





Reminder to use Scribblesheet &
Good luck with your midterm!